Clinical Study Summary Report (CSSR)
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1. Administrative & Study Identification

Full Study Title: A Phase 1 Clinical Study to Investigate the Safety and Pharmacokinetics of MK-5684 in Japanese Participants with Metastatic Castration-resistant Prostate Cancer (mCRPC)
Short Title/Acronym: MK-5684-005 (Opevesostat mCRPC Phase 1)
Protocol Number: MK-5684-005
NCT Number: NCT06104449
Sponsor Name: Merck **Company:** Merck
Investigational Product: Opevesostat (MK-5684 / ODM-208)
Phase of Development: PHASE1
Medical Monitor: Tomoko Fujita (MSD K.K., Contact: +81-3-6272-1957, msdjrct@msd.com)

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2. Study Synopsis & Rationale

2.1 Study Objective & Summary * **Summary:** The purpose of this study is to assess the efficacy and safety of opevesostat in the treatment of Japanese men with metastatic castration-resistant prostate cancer (mCRPC) previously treated with Next Generation Hormonal Agent (NHA) and taxane-based chemotherapy.

2.2 Background & Rationale To address the clinical need for novel therapies in mCRPC patients who have progressed on prior treatments. Opevesostat is a selective CYP11A1 inhibitor that entirely suppresses the production of all steroid hormones and precursors, effectively bypassing androgen receptor ligand-binding domain (AR-LBD) mutations. This study is conducted to establish the safety, PK profile, and preliminary efficacy of opevesostat in a Japanese demographic previously treated with Next Generation Hormonal Agents (NHAs) and taxane-based chemotherapy.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** N/A * **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) * **Status Name:** ACTIVE_NOT_RECRUITING * **Phase Criteria Requirement:** Trial must be in PHASE1 (Criteria Type: PHASE_REQUIREMENT, Phase ID: PHASE1)

3.2 Planned Enrollment & Duration * **Targeted Enrollment (N):** 6 * **Number of Sites:** Multiple sites in Japan * **Estimated Study Start:** 14-Nov-2023 * **Estimated Primary Completion:** 20-Feb-2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria * Has histologically- or cytologically-confirmed adenocarcinoma of the prostate without small cell histology * Has current evidence of metastatic disease documented by either bone lesions on bone scan and/or soft tissue disease by computed tomography/magnetic resonance imaging (CT/MRI) * Has ongoing androgen deprivation with serum testosterone <50 ng/dL (<1.7 nmol/L) * Participants receiving bone anti-resorptive therapy (including, but not limited to bisphosphonate or denosumab) must have been on stable doses for ≥4 weeks prior to the start of study intervention. * Has progressed on or after treatment with at least 1 line of NHAs in metastatic hormone-sensitive prostate cancer (mHSPC) or in castration-resistant prostate cancer (CRPC) for a minimum of 12 weeks (e.g. abiraterone, enzalutamide, darolutamide, apalutamide), and with at least 1 line of taxane-based chemotherapy in mHSPC or in CRPC, or ineligibility for chemotherapy * Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 within 10 days prior to allocation * If capable of producing sperm, participant must agree to the following during the study treatment period and for at least 7 days after the last dose of opevesostat: Refrain from donating sperm, plus EITHER be abstinent OR must agree to use male condom.

4.2 Exclusion Criteria * Has a history of pituitary dysfunction * Has brain metastases * History of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 3 years * Has an active or uncontrolled autoimmune disease that has required systemic treatment in past 2 years (ie, with use of disease modifying agents, corticosteroids, or immunosuppressive drugs) * Has an active infection or other medical condition that would make corticosteroid contraindicated * Has serious persistent infection within 2 weeks prior to the start of the study intervention * Participants on an unstable dose of thyroid hormone therapy within 6 months prior to the start of the study intervention * Has poorly controlled diabetes mellitus * Hypotension: systolic blood pressure (BP) < 110 mmHg, or uncontrolled hypertension: systolic BP ≥ 160 mmHg or diastolic BP ≥ 95 mmHg, in 2 out of 3 recordings with optimized antihypertensive therapy * Has active or unstable cardio/cerebro-vascular disease, including thromboembolic event * Is unable to swallow orally administered medication or known gastrointestinal (GI) disease or GI procedure that may interfere with absorption of study intervention * Has undergone major surgery including local prostate intervention (excluding prostate biopsy) within 28 days prior to the start of the study intervention and not adequately recovered from the toxicities and/or complications * Has received aldosterone antagonist (e.g. spironolactone, eplerenone) and phenytoin within 4 weeks prior to the start of the study intervention * Has received radiotherapy within 4 weeks prior to the start of the study intervention, or radiation related toxicities, requiring corticosteroids * Has received chemotherapy within the last 4 weeks (2 weeks for oral or weekly chemotherapy; 6 weeks for nitrosoureas and mitomycin C) prior to the start of the study intervention * Has received prior enzalutamide and apalutamide within 3 weeks, or abiraterone and darolutamide within 2 weeks prior to the start of the start of the study intervention * Systemic use of the following medications within 2 weeks prior to the start of study intervention: strong cytochrome P450 (CYP)3A4 inducers: e.g., carbamazepine, rifampicin, phenobarbital, phenytoin, St John's Wort) and strong CYP3A4 inhibitors: e.g., itraconazole, ketoconazole, posaconazole, voriconazole, clarithromycin, telithromycin, grapefruit juice * Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Administration of killed vaccines are allowed. * Has used herbal products that may have hormonal anti-prostate cancer activity and/or are known to decrease PSA levels (eg, saw palmetto) within 4 weeks prior to the start of the study intervention * Has received treatment with 5-α reductase inhibitors (eg, finasteride or dutasteride), estrogens, and/or cyproterone within 4 weeks prior to the start of the study intervention * Has received an investigational agent or has used an investigational device within 4 weeks prior to study intervention administration * History of human immunodeficiency virus (HIV) infection * Has a history of Hepatitis B or active Hepatitis C virus * Has a "superscan" bone scan * Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy (in dosing exceeding 10 mg daily of prednisone equivalent) or any other form of immunosuppressive therapy within 7 days prior to the start of the study intervention

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** MK-5684 5 mg twice daily (BID), co-administered with hormone replacement therapy consisting of dexamethasone 1.5 mg once daily (QD) and fludrocortisone acetate 0.1 mg QD. * **Route of Administration:** Oral (Tablets) * **Duration of Treatment:** Continuously until documented disease progression, unacceptable adverse events (AEs), or other discontinuation criteria are met.

5.2 Pharmacological Tracked Data (Associated Therapy / Reference Class) * **Drug Name:** cortisone (Preferred Name: cortisone) * **ATC Code:** H02AB10 * **ATC Codes List:** H02AB10, S01BA03 * **Trade Names:** Cortone, Cortone acetate * **Source Level:** 5 * **RxNorm Code:** *None provided* * **EMA URL:** *None provided* * **Drug Semantic Type:** *None provided*

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent,

Medical History, Baseline Imaging, Labs | | **Baseline** | Day 1 (Cycle 1) | First Dose of Study Drugs, Baseline PK Sampling | | **Interim** | Regular Cycles | Safety Labs, PK Sampling, PSA Assessments, AE check | | **End of Study** | Follow-up | Discontinuation Visit, Final Survival Contact |

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7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Number of Participants Who Experience a Dose-limiting Toxicity (DLT) as Assessed Using Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0 by the Investigator:** The following events, if considered drug related by the Investigator, will be considered a DLT: Grade 4 hematologic toxicity lasting ≥7 days, except anemia and thrombocytopenia; Grade 3 nausea, vomiting, diarrhea or fatigue lasting >3 days despite optimal supportive care; other nonhematologic grade ≥3 toxicities of any duration (not laboratory); Grade ≥3 nonhematologic laboratory abnormality (if certain criteria are met); febrile neutropenia Grade 3 or Grade 4; missing >25% of opevesostat doses as a result of drug-related AE(s) during the first 28 days; Grade 5 toxicity. The number of participants who experience a DLT will be presented. 2. **Number of Participants Who Experience an Adverse Event (AE):** An AE is defined as any untoward medical occurrence associated with the use of a drug in a participant, whether or not considered drug related. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of a medicinal product and does not imply any judgment about causality. 3. **Number of Participants Who Discontinue Study Intervention Due to an AE:** An AE is defined as any untoward medical occurrence associated with the use of a drug in a participant, whether or not considered drug related. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of a medicinal product and does not imply any judgment about causality.

7.2 Secondary Outcomes 1. **Maximum Plasma Concentration (Cmax) of opevesostat** 2. **Time to Maximum Plasma Concentration (Tmax) of opevesostat** 3. **Area Under the Curve from Time 0 to 12 hours postdose (AUC0-12) of opevesostat**

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring of any untoward medical occurrence associated with the drug via clinical assessments and laboratory tests.
- **Serious Adverse Events (SAEs):** Reported per standard protocol safety reporting requirements, typically within 24 hours of investigator awareness.

- **Data Monitoring Committee (DMC):** Supervised internally by the Sponsor (Merck).

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Population (all participants who receive ≥ 1 dose of study drug); PK Analysis Population.
- **Sample Size Justification:** Target $N=6$ is adequate for a regional Phase 1 PK, tolerability, and safety bridging study in Japan.
- **Handling of Missing Data:** Standard Phase 1 conventions; generally, missing safety and PK data are not imputed.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Routine remote and onsite monitoring visits implemented by the sponsor to ensure adherence to protocol.
- **Audit Trail:** Standard EDC database locking and data clarification form mechanisms employed for data integrity.

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11. Study Identifiers & Governance

- **PostingID:** 319383341948819610
- **Primary ID:** MK-5684-005
- **Registry Number:** NCT06104449
- **Sponsor/Lead Organization:** Merck
- **Company:** Merck
- **Study Title:** A Study of Opevesostat (MK-568)4 in Japanese Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC) (MK-5684-005)
- **Official Regulatory Title:** A Phase 1 Clinical Study to Investigate the Safety and Pharmacokinetics of MK-5684 in Japanese Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC)

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12. Clinical Framework (Trial Specific)

- **Primary Condition / Name:** Neoplasms (Preferred Name: Neoplasms)
- **Disease Areas:** Prostatic Neoplasms, Metastatic Castration-Resistant Prostate Cancer
- **Preferred UMLS Name:** Metastatic Castration-Resistant Prostate Cancer
- **Semantic Type:** Neoplastic Process

- **Definition:** New abnormal growth of tissue. Malignant neoplasms show a greater degree of anaplasia and have the properties of invasion and metastasis, compared to benign neoplasms.
- **Ontology Coding:**
- **MeSH Code:** D009369
- **HPO Codes:** HP:0002664
- **SNOMED ID:** 108369006
- **SNOMED Hierarchy:** 400177003 | Neoplasm and/or hamartoma | Neoplasm and/or hamartoma (morphologic abnormality); 416939005 | Proliferative mass | Proliferative mass (morphologic abnormality); 4147007 | Mass | Mass (morphologic abnormality); 52988006 | Lesion | Lesion (morphologic abnormality); 49755003 | Morphologically abnormal structure | Morphologically abnormal structure (morphologic abnormality); 118956008 | Morphologically altered structure | Body structure, altered from its original anatomical structure (morphologic abnormality); 123037004 | Body structure | Body structure (body structure); 30217000 | Proliferation | Proliferation (morphologic abnormality); 57697001 | Growth alteration | Growth alteration (morphologic abnormality)
- **Medical Methodology:** CYP11A1 inhibition with concurrent glucocorticoid and mineralocorticoid replacement therapy

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13. Study Procedures & Methodology

- **Management Pathway:** Standard oncological pathway for mCRPC with targeted oral therapy
- **Education Protocol (HCP):** Identification and management of adrenal insufficiency / proper use of rescue cortisone/ hydrocortisone
- **Education Protocol (Patient):** Adherence to complex oral regimen and symptom monitoring for adrenal insufficiency
- **Quality Audit Mechanism:** Central laboratory PSA tracking and routine safety/PK data monitoring

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14. Observation & Follow-Up Schedule

- **Total Duration:** Until disease progression, plus final survival follow-up
- **Onsite Visit Cadence:** Regular treatment cycle visits (e.g., C1D1, continuous per protocol)
- **Remote Monitoring Frequency:** Follow-up contacts post-treatment discontinuation
- **Checklist Review Interval:** Routine reviews prior to each cycle administration

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Insert Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Insert Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]